

Transcriptome Demo

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Load required packages (you might have to figure out how to install some of these first...)

```
library(ballgown)
library(RColorBrewer)
library(genefilter)
library(dplyr)
library(devtools)
```

What is this code doing? It is creating a simple table mapping sample IDs to their biological stage.

```
pheno_data<-data.frame(ids = c("plank01", "plank02", "biofilm01", "biofilm02"),
                       stage = c("planktonic", "planktonic", "biofilm", "biofilm"))
```

import Ballgown object and check transcript number

```
bg<-readRDS("bg.RDS")
bg
```

```
## ballgown instance with 5737 transcripts and 4 samples
```

<what is this code doing?> Filtering to keep transcripts that vary across my samples

```
bg_filt = subset(bg, "rowVars(texpr(bg)) >1", genomesubset=TRUE)
bg_filt
```

```
## ballgown instance with 5163 transcripts and 4 samples
```

create a table of transcripts

```
results_transcripts<- stattest(bg_filt, feature = "transcript", covariate = "stage",
getFC = TRUE, meas = "FPKM")
results_transcripts<-data.frame(geneNames=geneNames(bg_filt),
transcriptNames=transcriptNames(bg_filt), results_transcripts)
```

choose a transcript to examine more closely (this is a demo, you need to choose another)

```
results_transcripts[results_transcripts$transcriptNames == "gene-PA3059", ]

##      geneNames transcriptNames  feature  id      fc      pval
## 3129      pelF      gene-PA3059 transcript 3129 0.001687425 0.002583846
##      qval
## 3129 0.9471885
```

what information are you given about this transcript?

We get expression levels Variance Significant stats Transcript metadata

This code looks at the results and picks rows where the p-value is less than 0.05

```
sigdiff <- results_transcripts %>% filter(pval<0.05)
dim(sigdiff)

## [1] 207  7
```

organize the table <by what metrics is the table being organized?>

It sorts by p-value and fold change

```
o = order(sigdiff[, "pval"], -abs(sigdiff[, "fc"]), decreasing=FALSE)
output = sigdiff[o, c("geneNames", "transcriptNames", "id", "fc", "pval", "qval")]
write.table(output, file="SigDiff.txt", sep="\t", row.names=FALSE, quote=FALSE)
head(output)

##      geneNames transcriptNames  id      fc      pval      qval
## 4091      .      gene-PA3992 4091 9.886091e+01 0.0003032315 0.9471885
## 4958      .      gene-PA4804 4958 3.563696e-04 0.0006661432 0.9471885
## 2745      .      gene-PA2690 2745 5.783390e-02 0.0014192618 0.9471885
## 2896      tpm      gene-PA2832 2896 1.786570e+03 0.0023414834 0.9471885
## 370      .      gene-PA0365 370 3.964652e-07 0.0023906201 0.9471885
## 3129      pelF      gene-PA3059 3129 1.687425e-03 0.0025838457 0.9471885
```

load gene names

```
bg_table = texpr(bg_filt, 'all')
bg_gene_names = unique(bg_table[, 9:10])
```

pull out gene expression data and visualize

```
gene_expression = as.data.frame(gexpr(bg_filt))
head(gene_expression)

##      FPKM.plank01 FPKM.plank02 FPKM.biofilm01 FPKM.biofilm02
## .      1.198359      0.9103059      2.526183      2.685373
```

```
## MSTRG.1      405.892761  400.8589780    232.324417    181.932617
## MSTRG.10     89.649139   78.5762100    35.010487    59.757320
## MSTRG.100    116.443428  106.2109530    92.206810    95.322479
## MSTRG.1000   7.833186    5.5019700    15.717344    42.342495
## MSTRG.1001   6.845010    4.7381980    38.199095    89.078876
```

<what is this code doing?

This code changes the column names and displays the first few rows of the data set

```
colnames(gene_expression) <- c("plank01", "plank02", "biofilm01", "biofilm02")
head(gene_expression)
```

```
##           plank01    plank02 biofilm01 biofilm02
## .           1.198359  0.9103059   2.526183   2.685373
## MSTRG.1     405.892761 400.8589780 232.324417 181.932617
## MSTRG.10    89.649139  78.5762100  35.010487  59.757320
## MSTRG.100   116.443428 106.2109530  92.206810  95.322479
## MSTRG.1000  7.833186   5.5019700  15.717344  42.342495
## MSTRG.1001  6.845010   4.7381980  38.199095  89.078876
```

```
dim(gene_expression)
```

```
## [1] 4592    4
```

load the transcript to gene table and determine the number of transcripts and unique genes

```
transcript_gene_table = indexes(bg)$t2g
head(transcript_gene_table)
```

```
##   t_id   g_id
## 1     1 MSTRG.1
## 2     2 MSTRG.2
## 3     3 MSTRG.3
## 4     4 MSTRG.3
## 5     5 MSTRG.4
## 6     6 MSTRG.5
```

```
length(row.names(transcript_gene_table))
```

```
## [1] 5737
```

```
length(unique(transcript_gene_table[, "g_id"]))
```

```
## [1] 4715
```

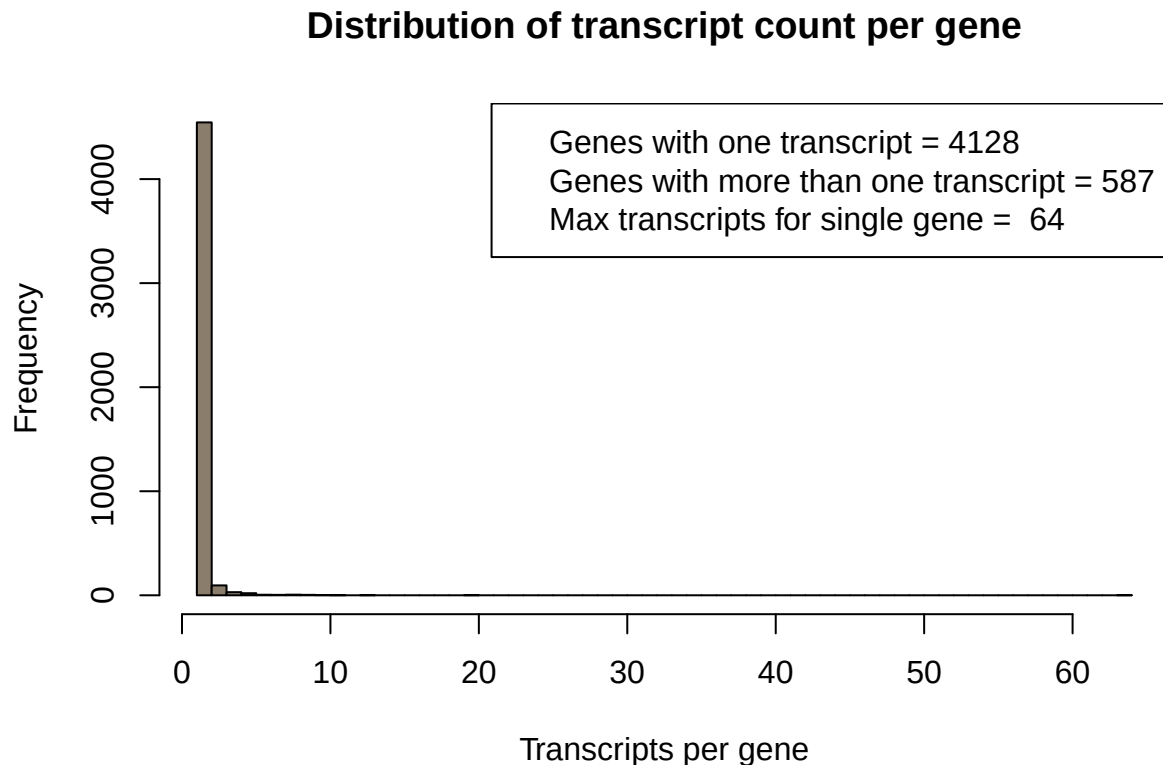
plot the number of transcripts per gene

```
counts=table(transcript_gene_table[, "g_id"])
c_one = length(which(counts == 1))
c_more_than_one = length(which(counts > 1))
c_max = max(counts)
hist(counts, breaks=50, col="bisque4", xlab="Transcripts per gene",
```

```

main="Distribution of transcript count per gene")
legend_text = c(paste("Genes with one transcript =", c_one),
paste("Genes with more than one transcript =", c_more_than_one),
paste("Max transcripts for single gene = ", c_max))
legend("topright", legend_text, lty=NULL)

```



<The Graph above is showing us Genes with one transcript, genes with more than one transcript and transcripts for one single gene >

create a plot of how similar the two replicates are for one another. We have two data sets... how can you modify this code in another chunk to create a plot of the other set?

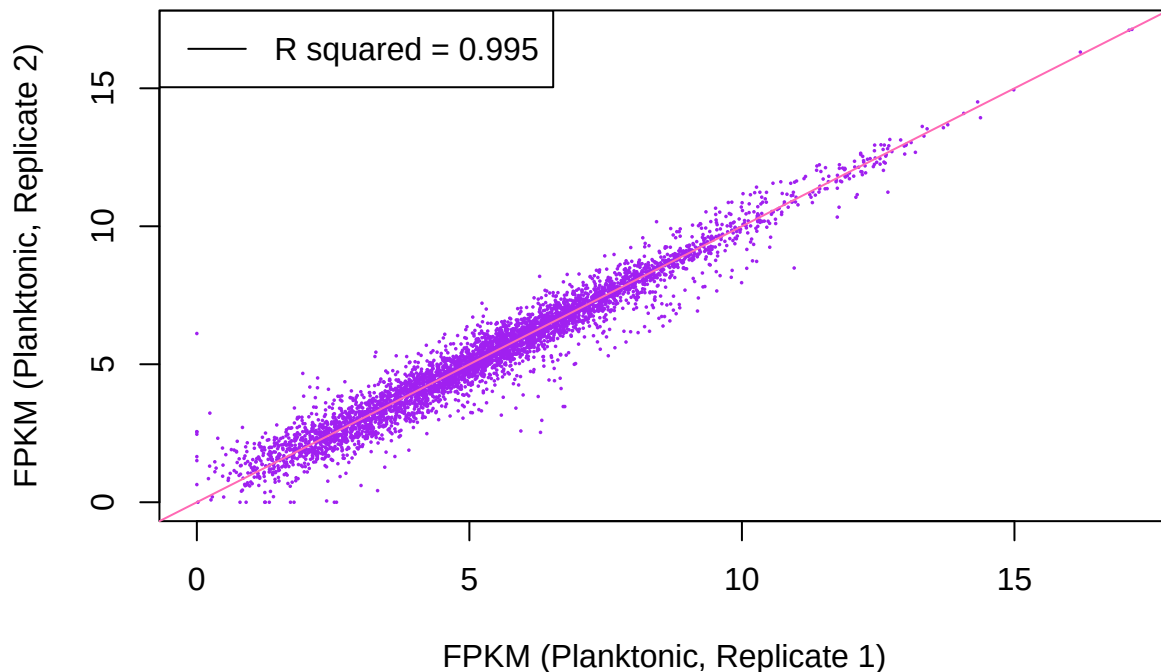
We can change the values we want to see for example if we want to zoom into one specific area of the graph.

```

x = gene_expression[, "plank01"]
y = gene_expression[, "plank02"]
min_nonzero=1
plot(x=log2(x+min_nonzero), y=log2(y+min_nonzero), pch=16, col="purple", cex=0.25,
xlab="FPKM (Planktonic, Replicate 1)", ylab="FPKM (Planktonic, Replicate 2)",
main="Comparison of expression values for a pair of replicates")
abline(a=0,b=1, col = "hotpink")
rs=cor(x,y)^2
legend("topleft", paste("R squared = ", round(rs, digits=3), sep=""), lwd=1, col="black")

```

Comparison of expression values for a pair of replicates



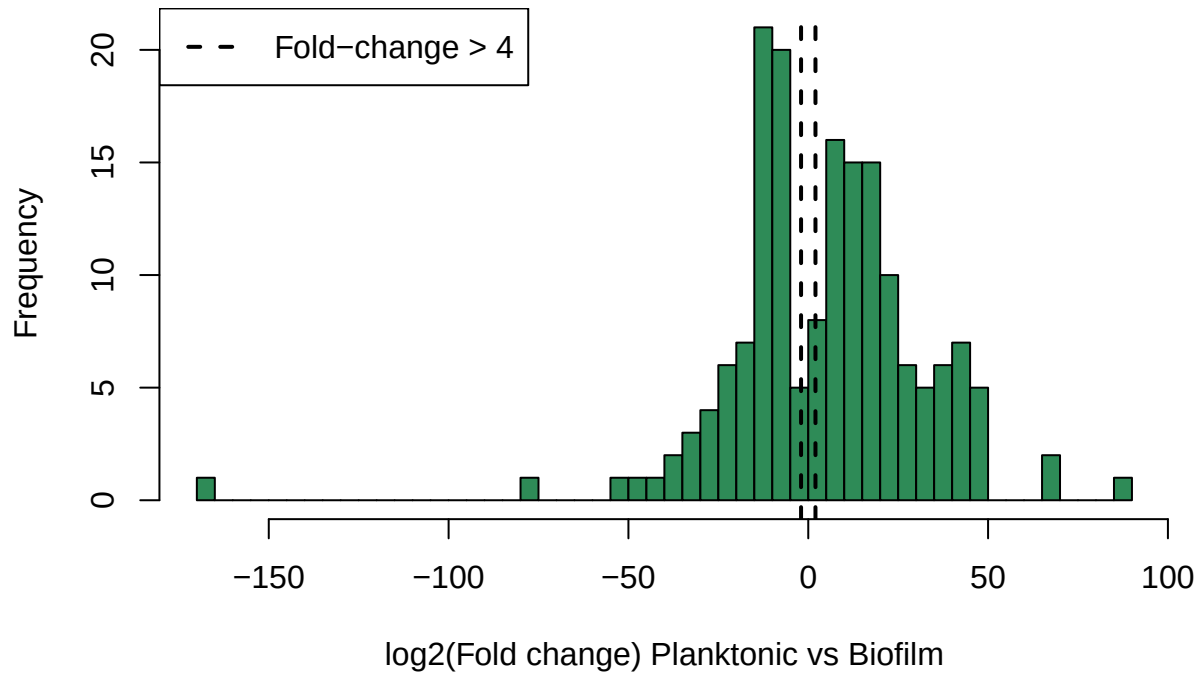
What does it mean if the two data sets are similar?

It means the data is related

create plot of differential gene expression between the conditions

```
results_genes = statstest(bg_filt, feature="gene", covariate="stage", getFC=TRUE, meas="FPKM")
results_genes = merge(results_genes, bg_gene_names, by.x=c("id"), by.y=c("gene_id"))
sig=which(results_genes$pval<0.05)
results_genes[, "de"] = log2(results_genes[, "fc"])
hist(results_genes[sig, "de"], breaks=50, col="seagreen",
xlab="log2(Fold change) Planktonic vs Biofilm",
main="Distribution of differential expression values")
abline(v=-2, col="black", lwd=2, lty=2)
abline(v=2, col="black", lwd=2, lty=2)
legend("topleft", "Fold-change > 4", lwd=2, lty=2)
```

Distribution of differential expression values

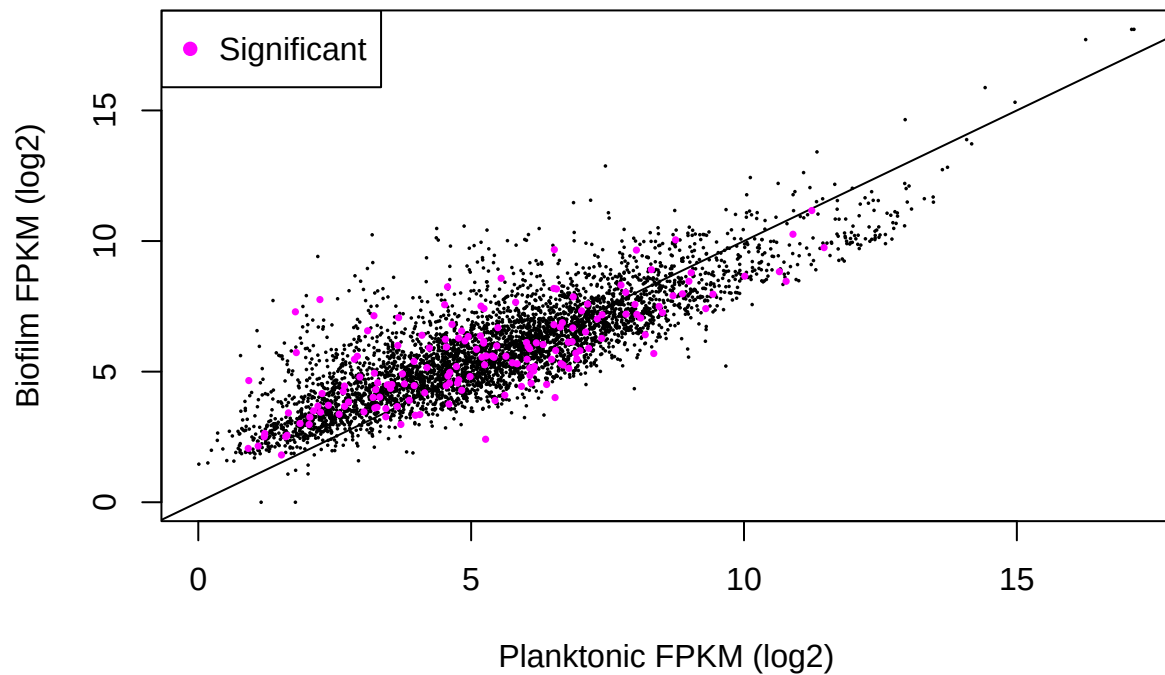


interpret the above figure: The figure above is giving us the differences in fold changes in planktonic vs. biofilm

Plot total gene expression highlighting differentially expressed genes

```
gene_expression[, "plank"] = apply(gene_expression[, c(1:2)], 1, mean)
gene_expression[, "biofilm"] = apply(gene_expression[, c(3:4)], 1, mean)
x = log2(gene_expression[, "plank"] + min_nonzero)
y = log2(gene_expression[, "biofilm"] + min_nonzero)
plot(x=x, y=y, pch=16, cex=0.25, xlab="Planktonic FPKM (log2)", ylab="Biofilm FPKM (log2)",
     main="Planktonic vs Biofilm FPKMs")
abline(a=0, b=1)
xsig=x[sig]
ysig=y[sig]
points(x=xsig, y=ysig, col="magenta", pch=16, cex=0.5)
legend("topleft", "Significant", col="magenta", pch=16)
```

Planktonic vs Biofilm FPKMs



make a table of FPKM values

```
fpkm = texpr(bg_filt, meas="FPKM")
```

choose a gene to determine individual expression (pick a different number than I did)

```
ballgown::transcriptNames(bg_filt)[2]
```

```
##          2  
## "gene-PA0002"
```

```
ballgown::geneNames(bg_filt)[2]
```

```
##      2  
## "dnaN"
```

transform to log2

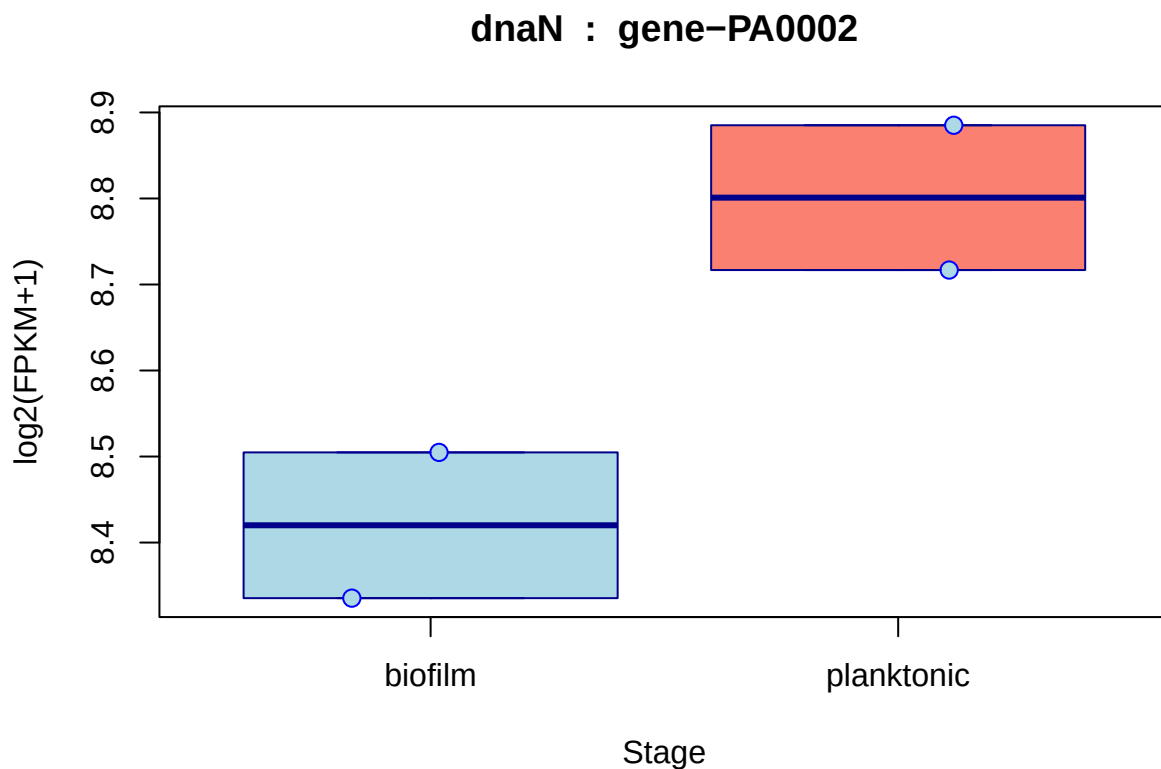
```
transformed_fpkm <- log2(fpkm[2, ] + 1)
```

make sure values are properly coded as numbers

```
numeric_stages <- as.numeric(factor(pheno_data$stage))  
jittered_stages <- jitter(numeric_stages)
```

plot expression of individual gene

```
boxplot(transformed_fpk ~ pheno_data$stage,  
  main=paste(ballgown::geneNames(bg_filt)[2], ' : ', ballgown::transcriptNames(bg_filt)[2]),  
  xlab="Stage",  
  ylab="log2(FPKM+1)",  
  col=c("lightblue", "salmon"),  
  border="darkblue")  
  
points(transformed_fpk ~ jittered_stages,  
  pch=21, col="blue", bg="lightblue", cex=1.2)
```



interpret the above figure: The figure above is showing us the difference in gene expression in biofilm vs planktonic